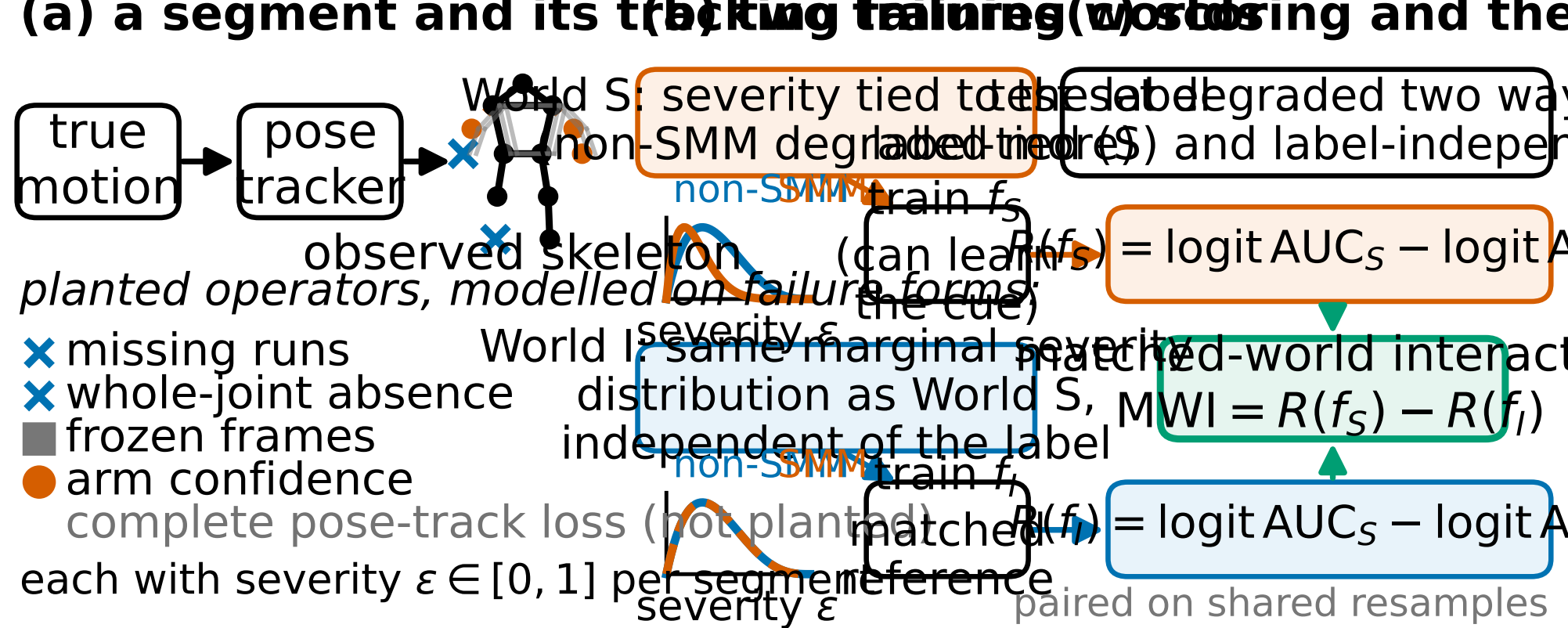
(a) a segment and its tracking failures
(b) the
(c) the
true motion
pose tracker
observed skeleton
missing runs
whole-joint absence
frozen frames
arm confidence
complete pose-track loss (not planted)
each with severity $\varepsilon \in [0, 1]$ per segment
World S: severity tied to the label
non-SMM degraded more
non-SMM
SMM
severity ε
train f_S
(can learn the cue)
World I: same marginal severity distribution as World S, independent of the label
non-SMM
SMM
severity ε
train f_I
matched reference
degraded two way
label-indepen
$R(f_S) = \text{logit AUC}_S - \text{logit A}$
matched-world interact
$\text{MWI} = R(f_S) - R(f_I)$
$R(f_I) = \text{logit AUC}_S - \text{logit A}$
paired on shared resamples